

QUESTION BANK

Maths — Algebra, Geometry & Trigonometry | 37 questions | source for Evalvia paper generator

Marks available by topic x difficulty

Topic	Easy	Medium	Hard	Total
Algebra	12	26	8	46
Geometry	12	26	21	59
Trigonometry	12	18	10	40
All topics	36	70	39	145

Note the thin cells: Algebra/hard and Trigonometry/hard each have a single question, and no hard-difficulty MCQs exist anywhere in the bank. A blueprint that leans on these cells is the case the generator's conflict-handling has to cover.

Algebra

ALG-1. Solve for x : $2x + 7 = 19$. [2 marks]

id: ALG-E1 | difficulty: easy | type: MCQ

- A. 4
- B. 6
- C. 8
- D. 12

ALG-2. Simplify: $3(x + 4) - 2x$. [2 marks]

id: ALG-E2 | difficulty: easy | type: MCQ

- A. $x + 12$
- B. $5x + 4$
- C. $x + 4$
- D. $5x + 12$

ALG-3. If $5x = 45$, find x . [2 marks]

id: ALG-E3 | difficulty: easy | type: MCQ

- A. 7
- B. 8
- C. 9
- D. 10

ALG-4. Evaluate $3x - 2$ when $x = 5$. [3 marks]

id: ALG-E4 | difficulty: easy | type: SHORT

ALG-5. Factorise: $x^2 + 5x$. [3 marks]

id: ALG-E5 | difficulty: easy | type: SHORT

ALG-6. Solve the quadratic: $x^2 - 5x + 6 = 0$. [3 marks]

id: ALG-M1 | difficulty: medium | type: MCQ

- A. $x = 1, 6$
- B. $x = 2, 3$
- C. $x = -2, -3$
- D. $x = 0, 5$

ALG-7. Solve the system: $2x + y = 7$, $x - y = 2$. [3 marks]

id: ALG-M2 | difficulty: medium | type: MCQ

- A. $x=3, y=1$
- B. $x=2, y=3$
- C. $x=1, y=5$
- D. $x=4, y=-1$

ALG-8. Simplify: $(x^2 - 9) / (x - 3)$. [4 marks]

id: ALG-M3 | difficulty: medium | type: SHORT

ALG-9. Solve for x: $3(2x - 1) = 2(x + 4)$. [4 marks]

id: ALG-M4 | difficulty: medium | type: SHORT

ALG-10. Find the roots of $x^2 + 2x - 8 = 0$ by factorisation. [4 marks]

id: ALG-M5 | difficulty: medium | type: SHORT

ALG-11. Derive the sum and product of roots of $ax^2 + bx + c = 0$ and verify for $x^2 - 7x + 12 = 0$. [8 marks]

id: ALG-M6 | difficulty: medium | type: LONG

ALG-12. Solve for real x: $x^4 - 5x^2 + 4 = 0$, and justify every root you discard. [8 marks]

id: ALG-H1 | difficulty: hard | type: LONG

Geometry

GEO-1. Find the area of a rectangle with sides 8 cm and 5 cm. [2 marks]

id: GEO-E1 | difficulty: easy | type: MCQ

- A. 13 cm^2
- B. 26 cm^2
- C. 40 cm^2
- D. 45 cm^2

GEO-2. What is the sum of interior angles of a triangle? [2 marks]

id: GEO-E2 | difficulty: easy | type: MCQ

- A. 90 deg
- B. 180 deg
- C. 270 deg
- D. 360 deg

GEO-3. Find the perimeter of a square with side 6 cm. [2 marks]

id: GEO-E3 | difficulty: easy | type: MCQ

- A. 12 cm
- B. 18 cm
- C. 24 cm
- D. 36 cm

GEO-4. Find the circumference of a circle with radius 7 cm (use $\pi = 22/7$). [3 marks]

id: GEO-E4 | difficulty: easy | type: SHORT

GEO-5. Find the area of a triangle with base 10 cm and height 6 cm. **[3 marks]**

id: GEO-E5 | difficulty: easy | type: SHORT

GEO-6. Find the area of a circle with radius 14 cm (use $\pi = 22/7$). **[3 marks]**

id: GEO-M1 | difficulty: medium | type: MCQ

- A. 308 cm^2
- B. 396 cm^2
- C. 616 cm^2
- D. 88 cm^2

GEO-7. A ladder 10 m long rests against a wall, foot 6 m from the wall. Find its height on the wall. **[3 marks]**

id: GEO-M2 | difficulty: medium | type: MCQ

- A. 6 m
- B. 7 m
- C. 8 m
- D. 9 m

GEO-8. Find the surface area of a cube with side 5 cm. **[4 marks]**

id: GEO-M3 | difficulty: medium | type: SHORT

GEO-9. Two similar triangles have sides in ratio 3:5. If the area of the smaller is 27 cm^2 , find the larger's area. **[4 marks]**

id: GEO-M4 | difficulty: medium | type: SHORT

GEO-10. Find the volume of a cylinder with radius 7 cm and height 10 cm (use $\pi = 22/7$). **[4 marks]**

id: GEO-M5 | difficulty: medium | type: SHORT

GEO-11. Prove that the diagonals of a parallelogram bisect each other, and verify using coordinates for A(0,0), B(4,0), C(6,3), D(2,3). **[8 marks]**

id: GEO-M6 | difficulty: medium | type: LONG

GEO-12. Prove the Pythagorean theorem using similar triangles. **[8 marks]**

id: GEO-H1 | difficulty: hard | type: LONG

GEO-13. A cone and a hemisphere share the same base radius r and the cone's height equals r . Derive the ratio of their volumes. **[8 marks]**

id: GEO-H2 | difficulty: hard | type: LONG

GEO-14. Find the shortest distance from the point (2, 3) to the line $3x + 4y - 12 = 0$, and derive the formula used. **[5 marks]**

id: GEO-H3 | difficulty: hard | type: SHORT

Trigonometry

TRI-1. Find $\sin(30^\circ)$. **[2 marks]**

id: TRI-E1 | difficulty: easy | type: MCQ

- A. 0
- B. $1/2$
- C. $\sqrt{2}/2$
- D. 1

TRI-2. Find $\cos(60^\circ)$. **[2 marks]**

id: TRI-E2 | difficulty: easy | type: MCQ

- A. $1/2$
- B. $\sqrt{3}/2$
- C. 1
- D. 0

TRI-3. Find $\tan(45^\circ)$. **[2 marks]**

id: TRI-E3 | difficulty: easy | type: MCQ

- A. 0
- B. $1/2$
- C. 1
- D. $\sqrt{3}$

TRI-4. State the value of $\sin^2(\theta) + \cos^2(\theta)$. **[3 marks]**

id: TRI-E4 | difficulty: easy | type: SHORT

TRI-5. Find the value of $\sin(90^\circ) - \cos(0^\circ)$. **[3 marks]**

id: TRI-E5 | difficulty: easy | type: SHORT

TRI-6. If $\sin(\theta) = 3/5$, find $\cos(\theta)$ for θ in the first quadrant. **[3 marks]**

id: TRI-M1 | difficulty: medium | type: MCQ

- A. $3/5$
- B. $4/5$
- C. $5/4$
- D. $5/3$

TRI-7. Simplify: $(1 - \cos^2 \theta) / \sin \theta$. **[3 marks]**

id: TRI-M2 | difficulty: medium | type: MCQ

- A. $\cos \theta$
- B. $\sin \theta$
- C. $\tan \theta$
- D. 1

TRI-8. A tower's shadow is 20 m when the sun's elevation is 45 deg. Find the tower's height. **[4 marks]**

id: TRI-M3 | difficulty: medium | type: SHORT

TRI-9. Prove: $(1 + \tan^2 \theta) = \sec^2 \theta$, using the identity $\sin^2 + \cos^2 = 1$. **[4 marks]**

id: TRI-M4 | difficulty: medium | type: SHORT

TRI-10. From a point 50 m from a building's base, the angle of elevation to the top is 30 deg. Find the height. **[4 marks]**

id: TRI-M5 | difficulty: medium | type: SHORT

TRI-11. A man standing on a 20 m tall building observes the angle of elevation of the top of a tower as 60 deg and the angle of depression of its foot as 30 deg. Find the height of the tower and the distance between the building and the tower. **[10 marks]**

id: TRI-L1 | difficulty: hard | type: LONG

Answer Key

ID	Topic	Diff.	Type	Marks	Answer
ALG-E1	Algebra	easy	mcq	2	6
ALG-E2	Algebra	easy	mcq	2	$x + 12$
ALG-E3	Algebra	easy	mcq	2	9
ALG-E4	Algebra	easy	short	3	13
ALG-E5	Algebra	easy	short	3	$x(x + 5)$
ALG-M1	Algebra	medium	mcq	3	$x = 2, 3$
ALG-M2	Algebra	medium	mcq	3	$x=3, y=1$
ALG-M3	Algebra	medium	short	4	$x + 3$, for $x \neq 3$
ALG-M4	Algebra	medium	short	4	$x = 11/4$
ALG-M5	Algebra	medium	short	4	$x = 2, x = -4$
ALG-M6	Algebra	medium	long	8	Sum = $-b/a = 7$, Product = $c/a = 12$; roots 3 and 4 satisfy both.
ALG-H1	Algebra	hard	long	8	Let $y = x^2$: $y^2 - 5y + 4 = 0$ $\Rightarrow y = 1, 4$ $\Rightarrow x = \pm 1, \pm 2$.
GEO-E1	Geometry	easy	mcq	2	40 cm^2
GEO-E2	Geometry	easy	mcq	2	180 deg
GEO-E3	Geometry	easy	mcq	2	24 cm
GEO-E4	Geometry	easy	short	3	44 cm
GEO-E5	Geometry	easy	short	3	30 cm^2
GEO-M1	Geometry	medium	mcq	3	616 cm^2
GEO-M2	Geometry	medium	mcq	3	8 m
GEO-M3	Geometry	medium	short	4	150 cm^2
GEO-M4	Geometry	medium	short	4	75 cm^2
GEO-M5	Geometry	medium	short	4	1540 cm^3
GEO-M6	Geometry	medium	long	8	Midpoint of AC = (3, 1.5) = midpoint of BD, confirming bisection.
GEO-H1	Geometry	hard	long	8	Drop an altitude from the right angle to the hypotenuse; the two smaller triangles are similar to the original triangle.
GEO-H2	Geometry	hard	long	8	$V_{\text{cone}} : V_{\text{hemisphere}} = (1/3 \pi r^3) : (2/3 \pi r^3) = 1:2$.
GEO-H3	Geometry	hard	short	5	$d = 3(2)+4(3)-12 /5 = 6/5 = 1.2$ units.
TRI-E1	Trigonometry	easy	mcq	2	$1/2$
TRI-E2	Trigonometry	easy	mcq	2	$1/2$
TRI-E3	Trigonometry	easy	mcq	2	1
TRI-E4	Trigonometry	easy	short	3	1
TRI-E5	Trigonometry	easy	short	3	0
TRI-M1	Trigonometry	medium	mcq	3	$4/5$
TRI-M2	Trigonometry	medium	mcq	3	$\sin \theta$

ID	Topic	Diff.	Type	Marks	Answer
TRI-M3	Trigonometry	medium	short	4	20 m
TRI-M4	Trigonometry	medium	short	4	Divide $\sin^2 + \cos^2 = 1$ by $\cos^2 \theta$ to get $\tan^2 + 1 = \sec^2$.
TRI-M5	Trigonometry	medium	short	4	Height = $50 \tan(30^\circ) \approx 28.9$ m
TRI-L1	Trigonometry	hard	long	10	Distance = $20/\tan(30^\circ) \approx 34.6$ m; tower height = $20 + 34.6 \tan(60^\circ) \approx 80$ m